

Francesco Di Filippo

Theoretical Gravitational Physics

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— Highlights

Fellowships and Awards: MSCA fellowship, FCT Fellowship (declined), Humboldt Fellowship, JSPS fellowship; Václav Votruba Prize, fifth prize 2024 Awards for Essays on Gravitation by the Gravity Research Foundation Awards. **Total ~ € 530.000.**

Strong Publication Record: Author of 38 publications in high-impact journals, ~2200 citations and h-index 23.

Event Organizer: Organizer of multiple events and workshops, including being one of the three founders of the Black Holes Inside and Out conference series, the largest conference devoted to black hole physics to date.

Invited Speaker and Panellist: Invited plenary speaker at 10 international conferences or workshops, selected plenary speaker at 1 international conference, invited panelist at 2 panel discussions, and invited to deliver several seminars in world-leading institutes.

— Academic positions

2025 – to date Humboldt Postdoctoral fellow at the Goethe University, Frankfurt, Germany.

2023 – 2025 Postdoctoral researcher at the Institute for Theoretical Physics, Charles University, Prague, Czechia.

2020 – 2023 JSPS postdoctoral researcher at the Yukawa Institute for Theoretical Physics, Kyoto, Japan.

— Education

2020 PhD in Astroparticle physics *cum laude*, SISSA International School for Advanced Studies, Trieste, Italy. Supervisor Prof. Stefano Liberati.

2016 Master's degree in physics with focus on Theoretical Physics *cum laude*, University of Salerno, Italy. Supervisors Prof. Carlos Barceló and Prof. Gaetano Lambiase.

— Fellowships and Awards

2025 Awarded the competitive MSCA postdoctoral fellowship. Approval rate ~15% (Fellowship accepted but not started yet) (~€ 200.000).

2025 Offered the competitive FCT fellowship. Approval rate ~14.6% (declined) (~€ 160.000).

2024 Awarded the competitive Humboldt fellowship (~€ 90.000).

2024 Fifth prize in the 2024 Awards for Essays on Gravitation by the Gravity Research Foundation. Co-writer of the essay “*Fully Extremal Black Holes: A Black Hole Graveyard?*” (Prize ~ \$ 400).

2021 Winner of the Václav Votruba Prize for the best thesis in theoretical physics (Prize ~ € 2.000).

2021 KAKENHI Grant-in-aid for scientific research (Co-Investigator). (~ € 11.000).

2021 Awarded the competitive JSPS postdoctoral fellowship. Approval rate ~11% (~€ 70.000).

2021 Honorable Mention in the 2021 Awards for Essays on Gravitation by the Gravity Research Foundation. Co-writer of the essay “*Hearts of Darkness: the inside out probing of black holes*”.

— Event organization

2025– to date Organizer of the Astrocoffee seminar series at the Institute for Theoretical Physics, Goethe University, Frankfurt, Germany.

- 11/2024 Organizer of the workshop “Towards a non-singular paradigm of Black Hole Physics” at IFPU in Trieste, Italy.
- 08/2024 Organizer of the “Black Holes Inside and Out 2024” conference at the Niels Bohr Institute in Copenhagen, Denmark.
- 08/2024 Member of the local organizing committee of the 14th conference on Relativistic Quantum Information in Prague, Czech Republic.
- 02/2024 Member of the local organizing committee of the PhD school “Measuring gravity”, Vietri sul mare (SA), Italy.
- 07/2023 Organizer of “Gravity 2023: dawn of field theoretic approaches” in Kyoto, Japan.
- 06/2022 Organizer of the conference “Gravity: Current challenges in black holes physics and cosmology” in Kyoto, Japan.
- 02/2022 Member of the local organizing committee for the YITP long-term workshop “Gravity and Cosmology 2024” Kyoto, Japan.
- 09/2021 Organizer of the “Black Holes Inside and Out 2021” online international conference.
- 03/2021 Member of the advisory board for the “Quantum Gravity, Higher Derivatives and Nonlocality” online conference.

— Habilitations for professorship

National Scientific Habilitation (“Abilitazione scientifica Nazionale”) for the position of Associate Professor in Theoretical Physics "Settore 02/A2" (08/06/2023– present). This qualification certifies that an individual possess all the requirements to be an associate professor in Italy.

— Teaching experience

- 2026 Tutor for the Advanced General Relativity course at Goethe University, Frankfurt, Germany.
- 2025 Invited lecturer of an intensive course “Advanced Topics in Black Hole Physics” held at the Institute for Basic Science in Daejeon, South Korea in November 2025.
- 2024 Invited lecturer on quantum black holes, at the summer school “*Towards Quantum Gravity*” in August 2024.
- 2015 - 2016 Teaching assistant for the Classical Mechanics course at the University of Salerno.

— Supervision experience

- 2025 - to date Co-supervisor (informal due to the Postdoc status) together with Prof. Luciano Rezzolla of the graduate student Daniel Jamplowski at the Goethe University, Frankfurt, Germany.
- 2022 Co-supervisor (informal due to the Postdoc status) together with Prof. Shinji Mukohyama of the master thesis of Takahiro Waki at the YITP, Kyoto University.

— Referee activity for journals

Referee for several journals of the field including Physical Review Letters, JCAP, Physical Review D, Physical Letters B, EPJC

— Selected oral Presentations

- 23/9/2025 **Invited talk** at the conference “*Crossroads in Strong Gravity*”, in Catania, Italy.
Towards a Non-singular Paradigm for Black Holes

- 27/06/2025 **Invited talk** at the conference “*From Puzzles to New Insights in Fundamental Physics*” in Campagna (SA), Italy.
Towards a Non-singular Paradigm for Black Holes
- 8/04/2025 **Invited talk** at the “Prague Spring 2025: CAS - IBS CTPU-CGA - ISCT Workshop in Cosmology, Gravitation and Particle Physics”, in Prague, Czechia.
Towards a non-singular paradigm for black hole physics
- 18/10/2024 **Invited talk** at the “CAS-IBS CTPU-CGA-Tokyo Tech 2024 Workshop”, in Toyohashi, Japan.
The end-point of gravitational collapse: Black holes or else?
- 23/10/2023 **Invited talk** at the conference “Puzzles in the Quantum Gravity Landscape: viewpoints from different approaches” at Perimeter Institute, Waterloo, Canada.
Hearts of Darkness: Nonsingular Black Holes Beyond General Relativity
- 01/09/2022 **Selected plenary talk** at the “Spanish and Portuguese relativity meeting (EREP) 2022”, Salamanca, Spain.
Viable regular black holes
- 03/08/2021 **Invited talk** at the Copernicus Webinar and Colloquium Series, online.
On the assumptions leading to the information loss paradox.
- 03/08/2021 **Invited talk** at the conference “Cosmology and Quantum Space Time”, Seoul, South Korea.
On the assumptions leading to the information loss problem.
- 28/04/2021 **Invited talk** at the conference “The Quantum and The Gravity”, Online.
Geodesically complete black holes: Possibilities and implications.
- 13/11/2019 **Invited talk** at “Autumn workshop on gravity and cosmology”, Warsaw, Poland.
On the uniqueness of general relativity.

— Panel discussions

- 26/10/2023 Invited panellist at the *black holes puzzles* panel during the conference “Puzzles in the Quantum Gravity Landscape: viewpoints from different approaches” at Perimeter Institute, Waterloo, Canada.
- 24/10/2023 Invited panellist at the *pizza & career* panel for young researcher during the conference “Puzzles in the Quantum Gravity Landscape: viewpoints from different approaches” at Perimeter Institute, Waterloo, Canada.

— Network of collaborators

Main collaborators include Raúl Carballo-Rubio (IAA Granada, Spain), Stefano Liberati (SISSA Trieste, Italy), Matt Visser (Wellington University, New Zealand), Luciano Rezzolla (Goethe University Frankfurt, Germany), Shinji Mukohyama (YITP Kyoto University, Japan), Costantino Pacilio (INFN Rome, Italy), José M.M. Senovilla (Basque University, Bilbao, Spain), David Kubiznak (Charles University Prague, Czechia).

— Citation metrics (according to the Inspire database, as of April 21st 2026.)

Publications: 38;	Citations: 2223;	H-index:23	(Including large collaboration papers)
Publications: 35;	Citations: 1621;	H-index:21	(Excluding large collaboration papers)

— Invited Seminars

I have been invited to deliver 20 seminars in world leading institutions including DAMPT (Cambridge, UK), Niels Bohr Institute (Copenhagen, Denmark), Johns Hopkins University (Baltimore, USA), Cornell University (Ithaca, USA), Tokyo Institute of Technology (Tokyo, Japan), Sapienza University (Rome, Italy).

— **Papers** (All papers are in alphabetical order)

- [31] R. Carballo-Rubio, F. Di Filippo, S. Mukohyama, K. Okabayashi,
Macroscopic backreaction of the trace anomaly on classical vacuum backgrounds
Preprint: arXiv 2512.10710
- [30] F. Di Filippo, S. Mukohyama, J. M. M. Senovilla,
On the regularity of deformed extremal horizons
Preprint: arXiv 2512.05200
- [29] F. Di Filippo,
Radiating black holes in general relativity need not be stable
Accepted for publication on Phys. Rev. Lett. Preprint: arXiv 2510.20649
- [28] R. Carballo-Rubio, F. Di Filippo, S. Mukohyama, K. Okabayashi,
Imprints of quantum vacuum fluctuations on the gravitational field of a spherical mass
JHEP 02 (2026) 129
- [27] L. Buoninfante, F. Di Filippo, I. Kolář, F. Saueressig,
Dust collapse and horizon formation in Quadratic Gravity
JCAP 01 (2025) 114
- [26] F. Di Filippo, L. Rezzolla,
Can light rings self-gravitate?
Phys. Rev. D 111, L021504
- [25] F. Di Filippo, S. Liberati, M. Visser,
Fully Extremal Black Holes: A Black Hole Graveyard.
International Journal of Modern Physics 10.1142/S0218271824400054
- [24] F. Di Filippo,
The nature of inner light-rings.
Phys.Rev.D 110 (2024) 8, 084026.
- [23] Francesco Di Filippo, Ivan Kolář, David Kubiznak,
Inner-extremal regular black holes from pure gravity.
Phys.Rev.D 111 (2025) 4, L041505.
- [22] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Mass inflation without Cauchy horizons.
Phys.Rev.Lett. 133 (2024) 18, 181402
- [21] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Constraints on thermalizing surfaces from infrared observations of supermassive black holes.
JCAP 11 (2023) 041.
- [20] F. Di Filippo, N. Ogawa, S. Mukohyama, Takahiro Waki,
Soft hair, dressed coordinates, and information loss paradox.
Phys.Rev.D 108 (2023) 4, 044034
- [19] R. Carballo-Rubio, F. Di Filippo, S. Liberati, C. Pacilio, M. Visser,
Comment on "Stability properties of Regular Black Holes".
Phys.Rev.D 108 (2023) 12, 128501.
- [18] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
A connection between regular black holes and horizonless ultracompact stars.
JHEP 08 (2023) 046

- [17] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
*Constraints on horizonless objects after the EHT observation of Sagittarius A**.
JCAP 08 (2022) 08, 055.
- [16] R. Carballo-Rubio, F. Di Filippo, S. Liberati, C. Pacilio, M. Visser,
Regular black holes without mass inflation instability.
JHEP 09 (2022) 118.
- [15] E. Berti, V. Cardoso, M.H.Y. Cheung, F. Di Filippo, F. Duque, P. Martens, S. Mukohyama,
Stability of the fundamental quasinormal mode in time-domain observations against small perturbations.
Phys.Rev.D 106 (2022) 8, 084011.
- [14] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Geodesically complete black holes in Lorentz-violating gravity.
JHEP 02 (2022) 122.
- [13] L. Buoninfante, Francesco Di Filippo, S. Mukohyama.
On the assumptions leading to the information loss paradox.
JHEP 10 (2021) 081.
- [12] R. Carballo-Rubio, F. Di Filippo, S. Liberati,
Hearts of Darkness: the inside out probing of black holes.
Int.J.Mod.Phys.D 30 (2021) 14, 2142024.
- [11] K. Aoki, F. Di Filippo, S. Mukohyama,
Non-uniqueness of massless transverse-traceless graviton.
JCAP 05 (2021) 071.
- [10] R. Carballo-Rubio, F. Di Filippo, S. Liberati, C. Pacilio M. Visser,
Inner horizon instability and the unstable cores of regular black holes.
JHEP 05 (2021) 132.
- [9] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Causal hierarchy in modified gravity.
JHEP 12 (2020) 055.
- [8] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Geodesically complete black holes.
Phys.Rev.D 101 (2020) 084047.
- [7] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Opening the Pandora's box at the core of black holes.
Class.Quant.Grav. 37 (2020) 14, 14.
- [6] R. Carballo-Rubio, F. Di Filippo, N. Moynihan,
Taming higher-derivative interactions and bootstrapping gravity with soft theorems.
JCAP 1910 (2019) 030.
- [5] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Phenomenological aspects of black holes beyond general relativity.
Phys. Rev., D98, 124009 (2018).
- [4] R. Carballo-Rubio, F. Di Filippo, S. Liberati, C. Pacilio, M. Visser,
On the viability of regular black holes.
JHEP 1807 023 (2018).

- [3] R. Carballo-Rubio, F. Di Filippo, S. Liberati,
Minimally modified theories of gravity: a playground for testing the uniqueness of general relativity.
JCAP 1806 no.06, 026, (2018).
- [2] C. Barceló, R. Carballo-Rubio, F. Di Filippo, L. J. Garay,
From physical symmetries to emergent gauge symmetries.
JHEP 1610 084 (2016).
- [1] F. Di Filippo, C. Noce,
Exact Electronic Bands for a Periodic Pöschl--Teller Potential.
Commun.Theor.Phys. 66 no.5, (2016).

— **Chapter in book** (Alphabetic order)

- [CB2] L. Buoninfante, F. Di Filippo,
Is the information loss problem a paradox?

Part of the book: The Black Hole Information Paradox.
Editors Ali Akil and Cosimo Bambi
- [CB1] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser,
Singularity-free gravitational collapse: From regular black holes to horizonless objects.
Part of the book: *Regular Black Holes Towards a New Paradigm of Gravitational Collapse.*
Editor C. Bambi.

— **Conference report** (Speaker first)

- [CR1] F. Di Filippo, R. Carballo-Rubio, S. Liberati, C. Pacilio, M. Visser,
On the Inner Horizon Instability of Non-Singular Black Holes.
Int.J.Mod.Phys.D 30 (2021) 14, 2142024.

— **Lecture notes** (Alphabetic order)

- [LN1] Ivano Basile, Luca Buoninfante, Francesco Di Filippo, Benjamin Knorr, Alessia Platania, Anna Tokareva,
Lectures in Quantum gravity.
Arxiv 2412.08690.

— **Large collaboration paper**

- [LC3] R. Carballo-Rubio, F. Di Filippo, S. Liberati, M. Visser, et al.
Towards a Non-singular Paradigm of Black Hole Physics
ArXiv: 2501.05505. To be published in JCAP.
- [LC2] Editors: Luca Buoninfante, Raúl Carballo-Rubio, Vitor Cardoso, Francesco Di Filippo, Astrid Eichhorn,
Black Holes Inside and Out 2024: visions for the future of black hole physics
ArXiv: 2410.14414
- [LC1] E. Barausse et al.,
Prospects for fundamental physics with LISA
General Relativity and Gravitation 52 (8), 1-33